

DP Computer Science: A 4.3.4 / 4.3.5 Unsupervised Learning: Clustering & Association

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Describe **unsupervised learning** as a paradigm that works with **unlabeled data** to find hidden patterns
 - Explain how **clustering algorithms**, such as **k-means**, are used to group similar data points together
 - Describe how **association rule mining** is used to discover relationships between items in large datasets
 - Analyse a given scenario and determine whether clustering or association rule mining is the more appropriate technique
 - Interpret a simple association rule and explain its meaning in a real-world context
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Key Vocabulary

Term	Definition
Unsupervised learning	A machine learning paradigm that finds hidden patterns in unlabeled data
Clustering	The task of grouping data points so that points in the same group (cluster) are more similar to each other than to those in other groups
k-means	A clustering algorithm that partitions data into k clusters based on distance to cluster centroids
Centroid	The centre point of a cluster, calculated as the mean of all points in the cluster

Term	Definition
Association rule mining	A technique to discover interesting relationships (if-then rules) between variables in large datasets
Antecedent	The "if" part of an association rule (the condition)
Consequent	The "then" part of an association rule (the result)
Market basket analysis	A common application of association rule mining to find products frequently bought together

Approaches to Learning (ATL) Elements

Skill	Description
Thinking (Pattern Recognition & Analysis)	Identifying inherent patterns in data without pre-existing labels; analysing relationships uncovered by association rules
Communication	Articulating the logic behind grouping items; explaining the meaning of an association rule clearly

2. Unsupervised Learning: The Big Picture

Supervised vs. Unsupervised Learning

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SUPERVISED vs. UNSUPERVISED LEARNING

SUPERVISED LEARNING

- Data has LABELS (answers)
- Goal: Predict labels for new data
- Examples: Linear Regression, K-NN, Decision Trees
- Question: "What is this?" (Classification)
or "How much?" (Regression)

UNSUPERVISED LEARNING

- Data has NO LABELS (no answers)
- Goal: Discover hidden patterns and structure
- Examples: Clustering, Association Rule Mining
- Question: "What patterns exist?"

The Hook: Supermarket Scenario

"A supermarket has data on millions of shopping baskets, but they don't have pre-defined 'types' of shoppers. How could they discover what kinds of customer groups they have?"

Answer: This is an **Unsupervised Learning** problem—we need to find hidden patterns without any labels.

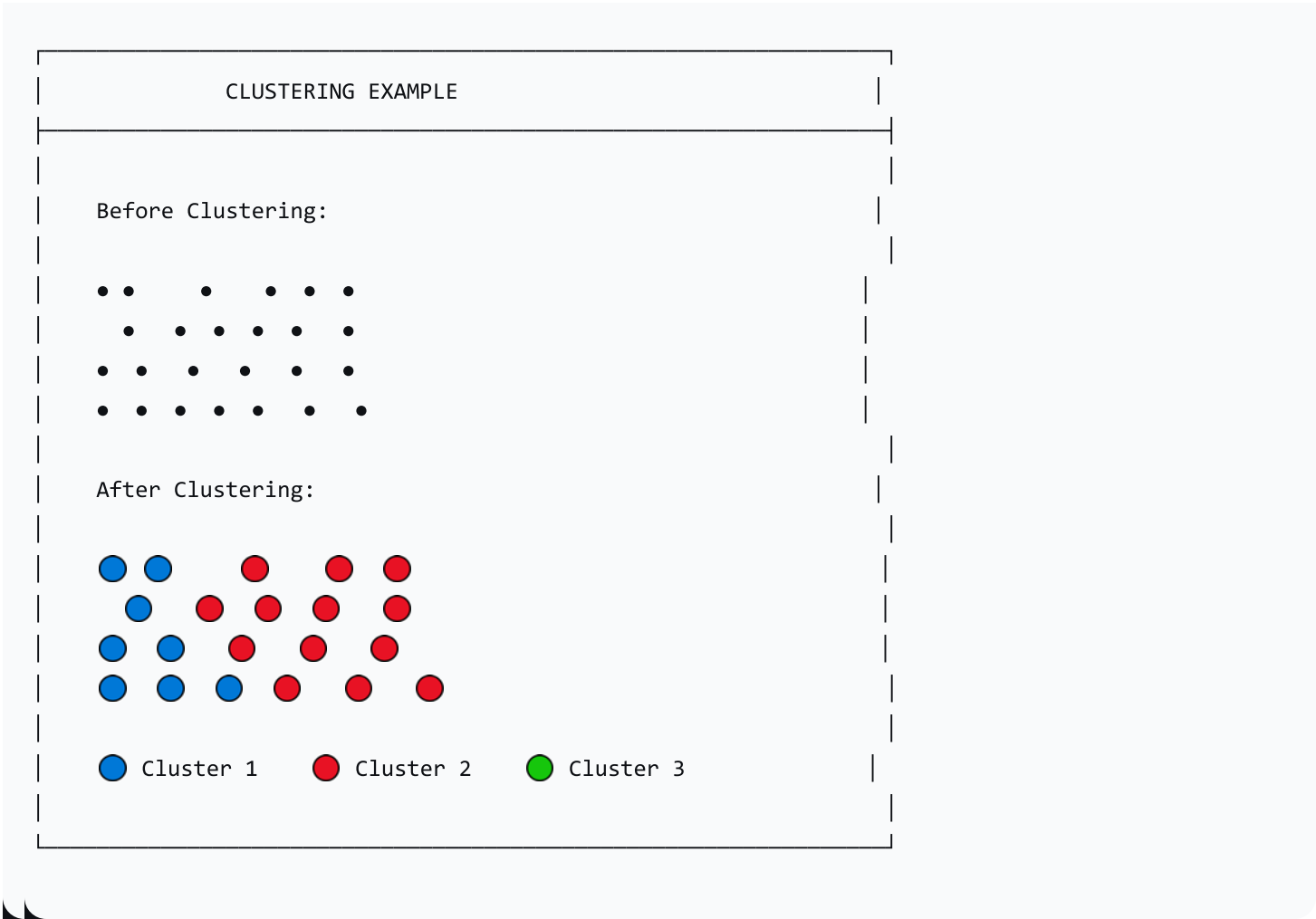
3. Clustering (A4.3.4)

What is Clustering?

Clustering is the task of grouping data points so that points in the same group (cluster) are more similar to each other than to those in other groups.

The Goal of Clustering

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k-means Algorithm

k-means is a clustering algorithm that partitions data into k clusters based on distance to cluster centroids.

How k-means Works

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K-MEANS ALGORITHM STEPS

Step 1: Choose k (number of clusters)

Step 2: Randomly place k centroids

Step 3: Assign each point to the nearest centroid

Step 4: Move each centroid to the mean of its cluster

Step 5: Repeat Steps 3-4 until clusters stabilise

Iteration 1:
(Initial)

Iteration 2:
(Update)

Iteration 3:
(Converged)



Visual Example of k-means

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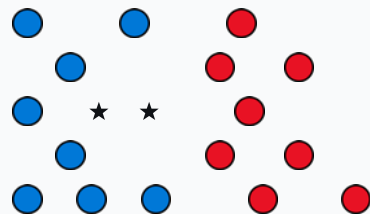
K-MEANS VISUAL EXAMPLE

Step 1: Choose $k = 2$



★ = Initial centroids

Step 2: Assign points to nearest centroid



● = Cluster 1 ● = Cluster 2

Step 3: Move centroids to cluster centres

(Repeat until clusters stabilise)

Real-World Applications of Clustering

Application	Description
Customer Segmentation	Group customers by purchasing behaviour for targeted marketing
Image Segmentation	Group pixels in an image to identify objects
Anomaly Detection	Identify unusual data points that don't belong to any cluster
Document Clustering	Group similar documents together
Social Network Analysis	Identify communities in social networks
Geographic Segmentation	Group locations with similar characteristics

4. Interactive Activity: "Human Clustering"

Instructions

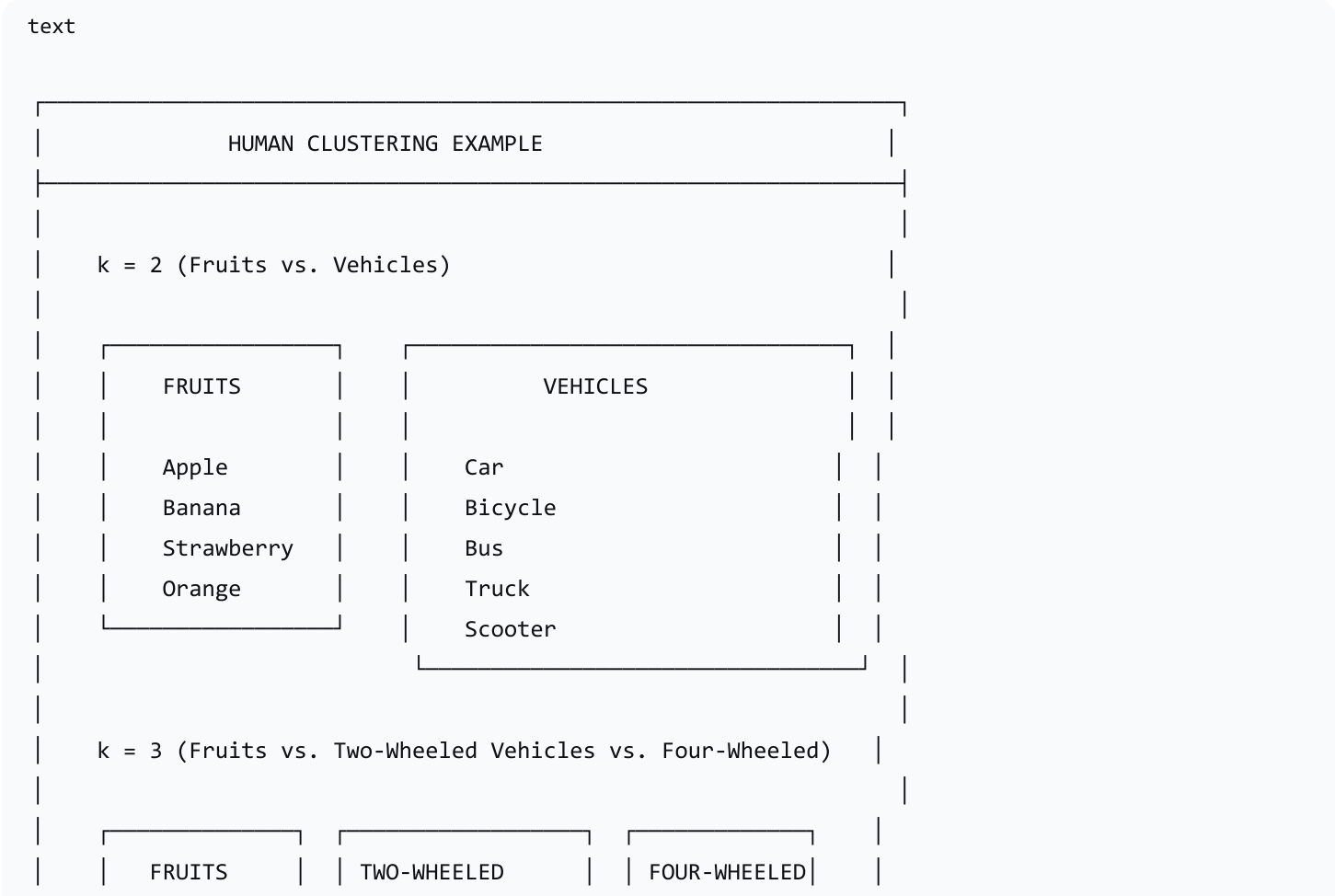
1. You are given a list of **10-12 diverse items**
2. If $k = 2$: How would you group these items? What are your two clusters?
3. If $k = 3$: How would you group these items? What are your three clusters?

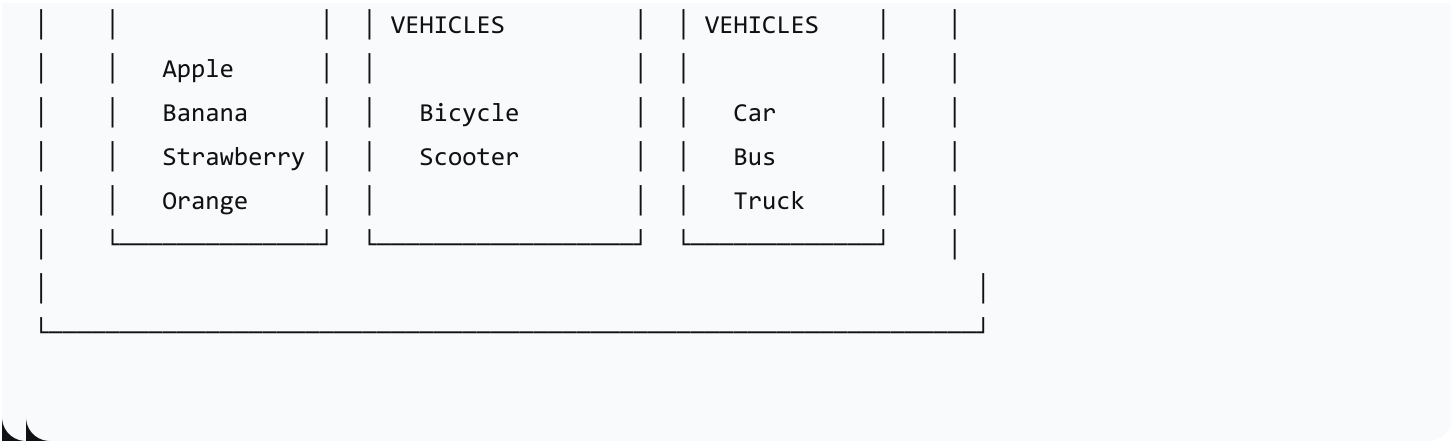
Example Item List

Item	Category
Apple	Fruit

Item	Category
Car	Vehicle
Banana	Fruit
Bicycle	Vehicle
Bus	Vehicle
Strawberry	Fruit
Orange	Fruit
Truck	Vehicle
Scooter	Vehicle

Possible Groupings





Discussion Points

Question	Purpose
"Why did you choose those groupings?"	Understand similarity
"What different features did you consider?"	Understand feature importance
"How does the choice of 'k' affect the outcome?"	Understand k-means parameter

5. Association Rule Mining (A4.3.5)

What is Association Rule Mining?

Association rule mining is a technique used to discover interesting relationships (if-then rules) between variables in large datasets.

The Classic Example: Diapers and Beer

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DIAPERS AND BEER EXAMPLE

Rule: {Diapers} → {Beer}

Meaning: Customers who buy diapers are also likely to

buy beer.

Why? (Often) Young fathers shopping late at night
→ buy diapers and pick up beer.

How it was discovered:

- Data: Millions of supermarket receipts
- Algorithm: "Find items that are frequently bought together"
- Result: Diaper sales and beer sales are correlated

Structure of an Association Rule

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ASSOCIATION RULE STRUCTURE

$\{\text{Antecedent}\} \rightarrow \{\text{Consequent}\}$

Antecedent = "If" (the condition)

Consequent = "Then" (the result)

Example: $\{\text{Bread}\} \rightarrow \{\text{Butter}\}$

Meaning: If a customer buys bread,
they are likely to buy butter.

Association Rule Example

Rule	Meaning	Real-World Application
{Bread} → {Butter}	Bread and butter are often bought together	Store layout: place them near each other
{Smartphone} → {Phone Case}	Phone buyers also buy cases	Recommendation: "Customers who bought this phone also bought a case"
{Milk} → {Cereal}	Milk and cereal are often bought together	Bundling: Offer a discount for buying together
{High Vandalism} → {Theft}	Areas with high vandalism often have theft	Predictive policing: allocate resources to high-risk areas

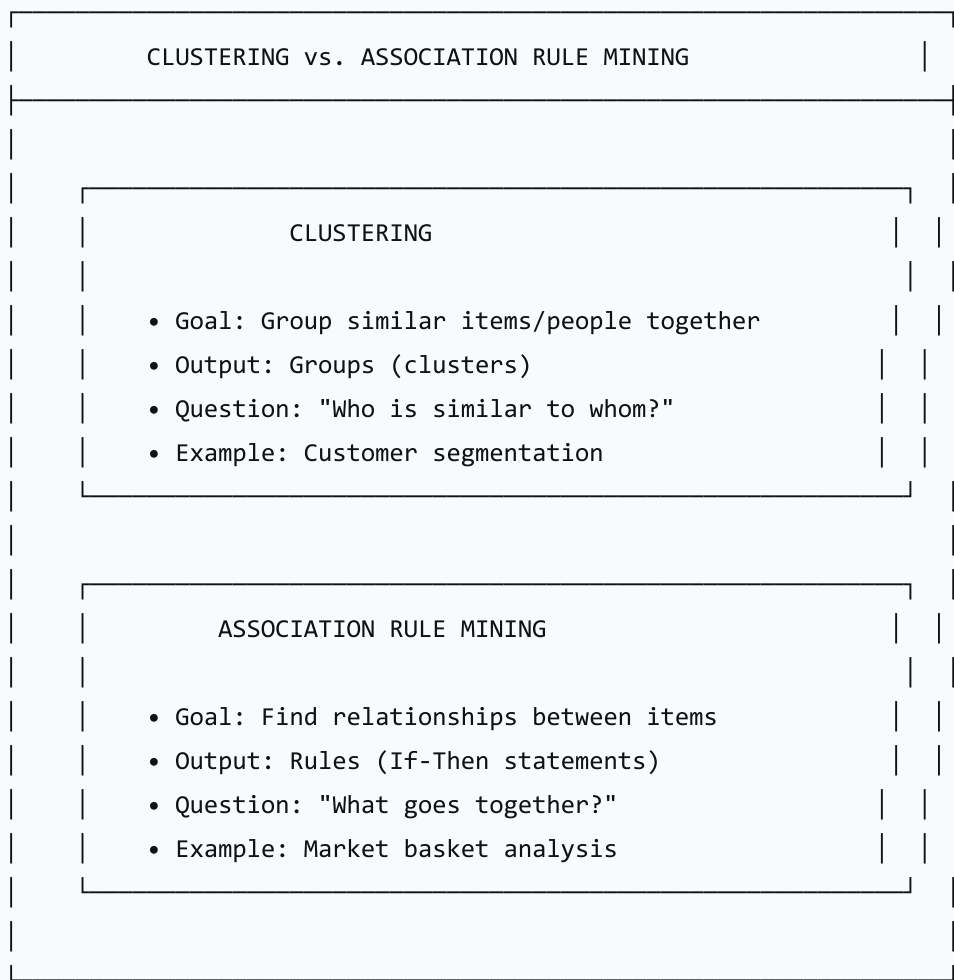
Applications of Association Rule Mining

Application	Description
Market Basket Analysis	Find products frequently bought together
Cross-selling	Recommend complementary products
Inventory Management	Understand which products are often purchased together
Crime Analysis	Find relationships between crime types (e.g., high vandalism → theft)
Web Recommendations	"Users who viewed this also viewed..."
Medical Diagnosis	Find symptoms that frequently occur together

6. Clustering vs. Association Rule Mining

Comparison

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When to Use Which?

Scenario	Technique	Why?
Group customers by shopping habits	Clustering	You want to find groups of similar customers
Find products that are often bought together	Association Rules	You want to find relationships between products
Segment images for object detection	Clustering	You want to group similar pixels/regions

Scenario	Technique	Why?
Find symptoms that frequently co-occur	Association Rules	You want to find relationships between symptoms
Identify communities in social networks	Clustering	You want to find groups of similar users
Recommend products based on purchase history	Association Rules	You want to find what products go together

Summary

Feature	Clustering	Association Rules
Goal	Group similar items	Find relationships between items
Output	Clusters (groups)	Rules (If-Then)
Question	"Who is similar?"	"What goes together?"
Example	Customer segments	Diapers → Beer

7. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Recap & Hook	10 min	Recap supervised learning; introduce unsupervised learning with supermarket scenario
Lecture: Clustering	15 min	Define clustering; k-means algorithm; real-world applications

Phase	Time	Activity
Interactive Activity	15 min	"Human Clustering" — group items with $k=2$ and $k=3$
Lecture: Association Rules	15 min	Define association rule mining; structure of rules; real-world applications
Consolidation	5 min	Compare clustering and association rules; preview next lesson

Discussion Questions

Question	Purpose
"What is the difference between supervised and unsupervised learning?"	Differentiate paradigms
"How does k-means work?"	Explain clustering algorithm
"What is an association rule?"	Understand rule structure
"When would you use clustering instead of association rules?"	Compare techniques
"What is market basket analysis?"	Understand real-world application

8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Human Clustering Activity	Assess understanding of clustering concepts

Assessment Type	Description
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is unsupervised learning?
2. How does the k-means algorithm work?
3. What is an association rule? Give an example.
4. What is the difference between clustering and association rule mining?

9. Differentiation

Level	Strategy
Support	Use clear examples and analogies; step through the "Human Clustering" activity carefully
Challenge	Explore other clustering algorithms (hierarchical, DBSCAN); research market basket analysis in detail

10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.3.4 & A4.3.5 with examples)
- **Worksheet:** Clustering and association rule scenarios
- **Worksheet Answers**
- **Quizlet** (A4.3.4/A4.3.5 vocabulary flashcards)

11. Summary: Key Takeaways

Concept	Key Point
Unsupervised Learning	Finds patterns in unlabeled data
Clustering	Groups similar items together
k-means	Iterative algorithm; assigns points to nearest centroid
Association Rules	Finds relationships between items
Rule Structure	{Antecedent} → {Consequent}
Clustering Output	Groups (who is similar)
Association Output	Rules (what goes together)

text

KEY REMINDERS

- ✅ Unsupervised = no labels, find patterns
- ✅ Clustering = group similar items
- ✅ k-means = choose k, assign, update centroids
- ✅ Association Rules = find relationships
- ✅ Rule = Antecedent → Consequent
- ✅ Clustering output = groups
- ✅ Association output = rules

"Unsupervised learning helps us discover hidden patterns without labels—whether it's finding groups of similar customers or discovering that certain products are often bought together."